Finite Distinction Carriers and History Escape

Subtitle: Observational equivalence and minimal deterministic quotients

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Abstract

For a finite deterministic system with a fixed observation map, two states may have the same present observation but different observations after a sequence of updates. We characterize this history-dependent distinction by a descending sequence of equivalence relations. The recursive refinement equals comparison over all input words up to a given length. It stabilizes within the number of states minus the number of initial observation classes; every pair removed at a refinement step has a shortest separating word of the corresponding length. The stabilized relation is the largest observation-preserving update congruence, and its quotient is the smallest deterministic quotient preserving every state's future observations. We also give a sufficient report-factorization condition, its proof, and a counterexample when reports introduce an additional observation. The arguments, including empty-state and empty-alphabet cases, are written in the text. In this package only Theorem WD2-T1 is a Lean 4 theorem. Lemma WD2-L1 and records WD2-T2 through WD2-T6 are finite-instance checks or prose arguments and are not labeled proved. The results are an exposition of finite-state observational equivalence and classical state minimization, not a new reconstruction of physical theory.

1. Question and relation to state minimization

An initially hidden state difference can become visible after specified updates. This differs from processing an unchanged observation: the observer is allowed to update the state and then observe it again. The question is which initially indistinguishable pairs can be separated in this way, and how long a shortest separating sequence can be.

The model is a finite deterministic Moore machine with a finite output alphabet. Its refinement and quotient results belong to classical automata theory [1,2]. We claim no novelty for partition refinement, observational equivalence, or minimal deterministic quotients. The purpose is to state their hypotheses, witness order, and observation boundary in one finite formulation. The arguments below are self-contained; no unpublished companion result or unavailable proof-assistant module is required.

Package status (2026-10-09). Theorem WD2-T1 is elaborated in Lean 4. Reflexivity, symmetry, and transitivity are `Wd2.related_refl`, `Wd2.related_symm`, and `Wd2.related_trans`. Inclusion and stability are `Wd2.related_mono` and `Wd2.related_stable`. The five-state chain at horizon 3 is `Wd2.sharp_m5_bound`. The general class-count bound, the no-escape equivalence, the shortest-word construction, the quotient universal property, and the report-factorization argument are not

Lean theorems in this package. The Python checker evaluates the sharp chain, the update-stable pair, and the factored report.

2. Finite model and word convention

Fix finite sets S of states and Σ of input symbols, an integer $K \geq 0$, maps

$$u : S \times \Sigma \longrightarrow S, \qquad p : S \longrightarrow \{0,1\}^K,$$

and write $u_\sigma(s) = u(s,\sigma)$. The binary output representation is a choice, not an assertion that binary distinctions are primitive. The proofs also apply to a map into any finite output set.

Words are applied **left to right**. With ε denoting the empty word, define

$$u_\varepsilon = \mathrm{id}_S, \qquad u_{w\sigma} = u_\sigma \circ u_w.$$

Induction on the length of v gives $u_{wv} = u_v \circ u_w$, and in particular $u_{\sigma v} = u_v \circ u_\sigma$. This convention matters when updates do not commute. No order on the alphabet is assumed.

For a map $f : S \to Y$, its kernel relation is $\ker f = \{(s,t) : f(s) = f(t)\}$. An *update congruence* is an equivalence relation E on S such that sEt implies $u_\sigma(s) E u_\sigma(t)$ for every symbol σ. It preserves the declared observation when $E \subseteq \ker p$.

3. Finite-horizon observations

Set $R_0 = \ker p$ and, for integers $n \geq 0$, define

$$R_{n+1} = R_0 \cap \bigcap_{\sigma \in \Sigma} (u_\sigma \times u_\sigma)^{-1}(R_n).$$

The intersection over an empty alphabet is $S \times S$. An initially ambiguous pair $(s,t) \in R_0$ *escapes at depth* $n \geq 1$ if $(s,t) \in R_{n-1} \setminus R_n$. A *separating word* satisfies $p(u_w(s)) \neq p(u_w(t))$; an *escape witness* is a separating word of shortest length. "Shortest" does not mean a unique word. A tie-breaking choice would require an additional order on Σ.

Lemma WD2-L1 (word characterization). For every integer $n \geq 0$,

$$R_n = \bigcap_{|w| \leq n} \ker(p \circ u_w).$$

Proof. At $n = 0$ the only word is ε, so the formula is the definition of R_0. Every nonempty word of length at most $n+1$ has a first symbol and a suffix: $w = \sigma v$, $|v| \leq n$. By the induction hypothesis, equality of all its readouts is equivalent to $(u_\sigma(s), u_\sigma(t)) \in R_n$ for every σ. The empty word additionally requires $(s,t) \in R_0$. These are exactly the recursive conditions for R_{n+1}. □

4. Refinement and stabilization

Theorem WD2-T1 (refinement and fixed-point law). Every R_n is an equivalence relation and $R_{n+1} \subseteq R_n$. If $R_{n+1} = R_n$, then $R_m = R_n$ for every $m \geq n$.

Proof. Kernels of maps are equivalence relations, and intersections preserve that property. Lemma L1 therefore establishes equivalence. The set of words of length at most n is contained in the set for $n+1$, establishing the inclusion. Define

$$F(E) = R_0 \cap \bigcap_{\sigma \in \Sigma} (u_\sigma \times u_\sigma)^{-1}(E).$$

If $R_{n+1} = R_n$, then $F(R_n) = R_n$. Applying F repeatedly gives the claimed equality at all later indices. □

Classical record WD2-T2 (finite stabilization and a counting bound). Not a theorem of this package. Put $b_0 = |S/R_0|$. There is a least integer $N \geq 0$ such that $R_N = R_{N+1}$, and

$$N \leq |S| - b_0, \qquad R_\infty := \bigcap_{n \geq 0} R_n = R_N.$$

The number N counts strict refinement rounds. An implementation can need one additional comparison to establish $R_N = R_{N+1}$.

Proof. A strict inclusion of finite equivalence relations splits at least one class, increasing the number of classes by at least one. Starting with b_0 classes, no more than $|S| - b_0$ consecutive strict inclusions can occur. The first equality is permanent by T1. The descending intersection is therefore its stabilized value. If S is empty, both class counts and N are zero. If Σ is empty, the recursion is constant and $N = 0$. □

For sharpness when $m = |S| \geq 2$, take $S = \{0, \dots, m-1\}$, a single symbol a, $u_a(i) = \min(i+1, m-1)$, and $p(i) = 1$ precisely for $i = m-1$. At horizon $j \leq m-2$, the lowest class is $\{0, \dots, m-2-j\}$ and all larger states are singleton classes. Indeed state $i < m-1$ first reads one after exactly $m-1-i$ updates. Thus $b_0 = 2$ and $N = m-2 = |S| - b_0$ (including $N = 0$ when $m = 2$).

5. No escape and shortest witnesses

Classical record WD2-T3 (no-escape criterion). Not a theorem of this package. The following are equivalent:

1. R_0 is an update congruence;
2. $R_1 = R_0$;
3. $R_n = R_0$ for every $n \geq 0$;
4. no pair in R_0 has a separating word.

Proof. Since $R_1 \subseteq R_0$, equality holds exactly when every pair in R_0 is carried by every update back into R_0. This is condition 1. Condition 2 implies 3 by T1, and 3 implies 2 by taking $n = 1$. By L1, condition 3 says that every initially equivalent pair has equal readouts after every finite word, which is exactly condition 4. □

Classical record WD2-T4 (shortest separating-word construction). Not a theorem of this package. If $(s, t) \in R_n \setminus R_{n+1}$, then its shortest separating-word length is $n + 1$. Such a word can be constructed as a first symbol followed by a length-n suffix. Consequently every escaping pair has escape depth at most $N \leq |S| - b_0$.

Proof. Membership in R_n excludes separation by any word of length at most n, by L1. Failure of R_{n+1}, together with membership in R_0, provides a symbol σ for which the successor pair lies outside R_n. For $n = 0$, the empty suffix separates that successor pair. For $n \geq 1$, the recursion for membership of the original pair in R_n also places the successor pair in R_{n-1}. Induction supplies a shortest separating suffix v of length n. The word σv separates the original pair because $u_{\sigma v} = u_v \circ u_\sigma$. Its length is $n + 1$, which is minimal by the first observation. No pair leaves the sequence after its stabilization at R_N, giving the depth bound. □

6. The complete finite behavioral quotient

Classical record WD2-T6 (largest congruence and smallest deterministic quotient). Not a theorem of this package. The relation R_∞ is the largest update congruence contained in $\ker p$. The quotient $Q_* = S/R_\infty$ has well-defined updates and output

$$\bar{u}_\sigma([s]) = [u_\sigma(s)], \qquad \bar{p}([s]) = p(s).$$

Let $q : S \twoheadrightarrow Q$ be any surjection with deterministic maps $v_\sigma : Q \to Q$ and $o : Q \to \{0,1\}^K$ satisfying

$$q \circ u_\sigma = v_\sigma \circ q, \qquad p = o \circ q.$$

There is a unique surjection $h : Q \twoheadrightarrow Q_*$ such that $\pi = h \circ q$, where $\pi(s) = [s]$. It commutes with updates and preserves output. In particular $|Q| \geq |Q_*|$. Equality makes h an isomorphism of the deterministic output systems.

Proof. Stabilization and the recursion show that R_∞ is an update congruence contained in R_0. If $E \subseteq R_0$ is any update congruence, induction on word length shows that sEt implies $u_w(s)Eu_w(t)$ for every word. Their observations agree, and L1 gives $E \subseteq R_n$ for every n, hence $E \subseteq R_\infty$. This establishes maximality and also makes the displayed quotient maps well-defined.

For a specified quotient q, its kernel is an equivalence relation. The update equation makes it an update congruence, and the output equation places it inside R_0. Maximality yields $\ker q \subseteq R_\infty$. Define $h(q(s)) = [s]$. Kernel inclusion makes this independent of the choice of s, and surjectivity of q gives existence and uniqueness on all of Q. Every class $[s]$ lies in the image, so h is onto. Moreover,

$$h(v_\sigma(q(s))) = [u_\sigma(s)] = \bar{u}_\sigma(h(q(s))),$$

and $o(q(s)) = p(s) = \bar{p}(h(q(s)))$. This establishes compatibility. Finite surjectivity gives the cardinal bound; at equality h is bijective, and its inverse also respects the maps. □

Equivalently, the *behavior* of a state is the function $\beta_s : \Sigma^* \to \{0,1\}^K$ given by $\beta_s(w) = p(u_w(s))$. Lemma L1 identifies R_∞ with equality of behaviors. Thus Q_* classifies every state by all its future readouts, and the induced update sends a behavior β_s to $w \mapsto \beta_s(\sigma w)$. This completes the classification of states for the specified finite model. It does not classify physical theories or select quantum mechanics.

The cardinality assertion concerns quotients preserving readouts from **every** state in S. A model restricted to the states reachable from one initial state is a different minimization problem. If S is empty, a surjective q has empty codomain, and the statement still holds.

7. Reports are additional observations unless constrained

Fix a set Λ and a report $r : S \times \Sigma \to \Lambda$. For a word $w = \sigma_1 \cdots \sigma_m$, set $s_0 = s$ and $s_i = u_{\sigma_i}(s_{i-1})$. The report at step i is $r(s_{i-1}, \sigma_i)$: it is attached to the update from s_{i-1} to s_i. This specifies the timing convention.

Classical record WD2-T5 (sufficient report-factorization condition). Not a theorem of this package. Suppose there is a map $g : \{0,1\}^K \times \Sigma \times \{0,1\}^K \to \Lambda$ such that

$$r(s, \sigma) = g(p(s), \sigma, p(u_\sigma(s))).$$

If $(s,t) \in R_n$, then their report sequences agree along every common word of length at most n. Consequently adjoining these report sequences to all profile histories through horizon n leaves the relation R_n unchanged.

Proof. For each prefix of length $i \leq n$, L1 gives equality of the profile values reached from s and t. In particular the profiles before and after step i agree, and the input symbol is the same. Applying g therefore gives equal reports at that step. This holds at every step of every such word. The profile comparisons already impose R_n; the equal reports impose no further condition. □

This is sufficient, not a necessary characterization. At horizon infinity a report constant on R_∞-classes for each input can also preserve behavioral equivalence without factoring through just the two endpoint profiles. For example, let states x, y, z have profiles $0, 0, 1$, let a act identically, and let b send x to x and both y, z to z. Here R_∞ is equality. A report distinguishing x, y on input a preserves that equality relation but cannot factor through the identical endpoint profile pair $(0, 0)$ for a.

Without any report restriction even the one-step conclusion fails. Take $S = \{0, 1\}$, one identity update a, constant profile $p = 0$, and $r(s, a) = s$. Every profile history agrees, but the first reports differ. The report has supplied an additional observation, not recovered a difference from the unchanged profile.

8. Worked witnesses and boundary cases

Three-state escape. Let $S = \{x, y, z\}$, $\Sigma = \{a\}$, $p(x) = p(y) = 0$, $p(z) = 1$, and $u_a(x) = x$, $u_a(y) = u_a(z) = z$. Initially the classes are $\{x, y\}$ and $\{z\}$. One refinement makes every state a singleton; a is a shortest witness for x, y. Thus $N = 1 = 3 - 2$.

Noncommuting updates. Let $S = \{0, 1, 2, 3\}$, profiles $(0, 0, 0, 1)$, and specify the update values in state order by

$$u_a = (1, 2, 2, 3), \qquad u_b = (0, 0, 3, 3).$$

States $0, 1$ have equal readouts at the empty word and after either one-letter word. Under ab they reach $0, 3$, whose profiles differ. Under ba they both reach 1. Thus ab is a shortest separating word of length two, and reversing the word destroys this witness. In fact R_1 has classes $\{0, 1\}, \{2\}, \{3\}$ and R_2 is equality.

No symbols or no output bits. If Σ is empty, the only word is ε and no initially ambiguous pair escapes. If $K = 0$, the output set is a singleton, so all pairs have equal profiles forever, regardless of the updates. For empty S, every relation and quotient is empty and every assertion about a state pair is vacuous.

Mechanism inventory

Object	Mathematical role
Finite S	States being compared
Finite Σ and $u_\sigma : S \to S$	Permitted deterministic updates
Finite profile $p : S \to \{0, 1\}^K$	Declared observation
Equivalence relations R_n	Equality of readouts through horizon n
Separating word	Finite witness of observable difference

Object	Mathematical role
S/R_∞	Smallest deterministic quotient preserving all states' future readouts
Optional report r	Additional observation with explicit timing and factorization hypothesis
AuthorityEffect	None

Verification scope

Theorem WD2-T1 is elaborated in Lean 4 with no `sorry`. Lemma WD2-L1 and records WD2-T2 through WD2-T6 have prose arguments above. The Python checker evaluates finite instances of those statements; their general forms are not Lean theorems in this package. From the package root, this command checks pinned files, those finite tests, and the Lean build:

```
py -3.12 verify_deposit.py --lean --test code/tests/test_proved_rows.py
```

The reading notebook contains prose only and requires no kernel execution.

No cochain complex, Hilbert space, repair norm or physical probability rule is defined here. A quotient by behavioral equivalence is a quotient of a finite set, not a Banach quotient. Nothing here derives memory, agency, consciousness, or quantum theory from a deterministic update system.

References

1. L. Trevisan, *CS103: Mathematical Foundations of Computing*, Lecture 16, state minimization, Stanford University (2010). https://theory.stanford.edu/~trevisan/cs103-10/lecture16.pdf.
2. J. E. Hopcroft, R. Motwani, and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, third edition, Pearson (2007), Chapter 4. ISBN 978-0-321-45536-9.